



CYBERSECURITY INTERNSHIP PROGRAM

WEEK 2 · TASK 5

Networking Tools & Subnetting Basics

Practical Report: ping, traceroute, nslookup, netstat & CIDR subnetting

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PROGRAM	TechBiz Academy – Cybersecurity Track
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TryHackMe proof, command outputs & subnetting example enclosed.

1. Objective

This report documents the practical work completed for Week 2, Task 5 of the TechBiz Academy cybersecurity internship. The task focuses on foundational networking reconnaissance tools (ping, traceroute, nslookup and netstat) and basic subnetting concepts (network vs. host bits, CIDR notation). Mastery of these fundamentals is a prerequisite for the scanning and enumeration phases that follow later in the program.

The work is organised into three parts: (1) completion of the two mandatory free-tier TryHackMe rooms – **DNS in Detail** and **HTTP in Detail**; (2) hands-on execution and analysis of four core networking commands on a Windows PowerShell terminal; and (3) a solved subnetting example for a /24 CIDR block.

2. TryHackMe Rooms Completed (Proof of Completion)

Both mandatory free-tier rooms listed in the task brief were completed in full. Screenshots below show the completion summary generated by TryHackMe immediately after finishing each room, including tasks completed, points earned and the current streak.

2.1 DNS in Detail

This room covers how the Domain Name System resolves human-readable domain names into IP addresses, record types (A, AAAA, MX, NS, TXT, CNAME), and the recursive resolution process. Completed with 5 tasks finished and 112 points earned.

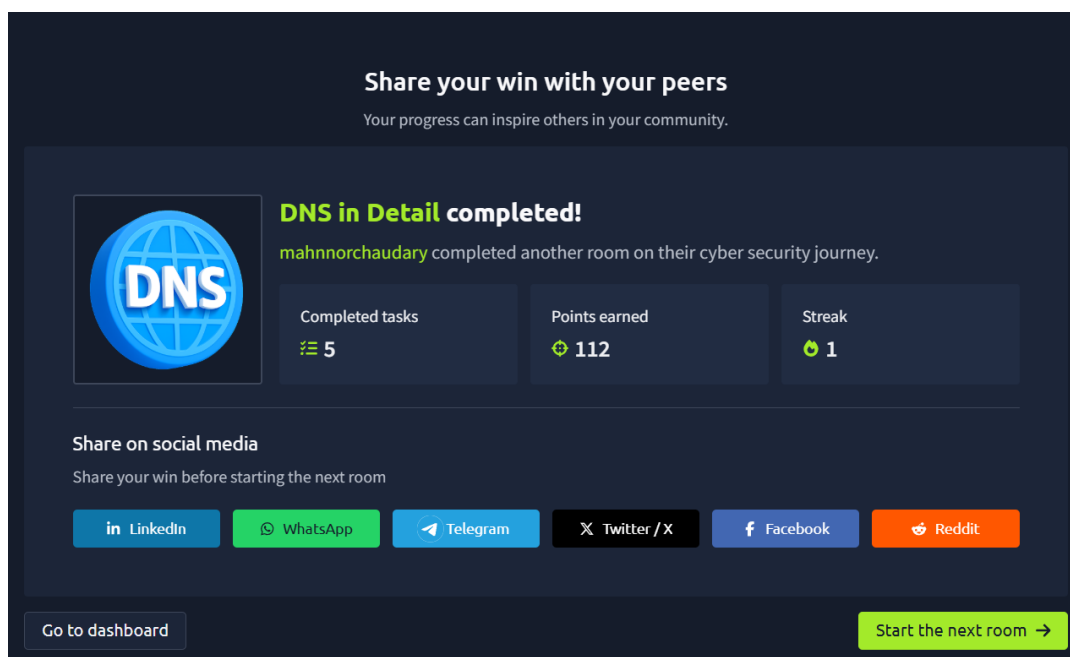
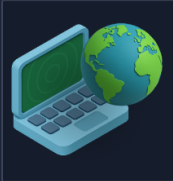


Figure 1 – TryHackMe: “DNS in Detail” room completion summary (5 tasks, 112 points).

2.2 HTTP in Detail

This room explains the HTTP request/response cycle, methods (GET, POST, etc.), status codes and headers – knowledge that underpins later web-application testing tasks. Completed with 7 tasks finished and 176 points earned.

Share your win with your peers
Your progress can inspire others in your community.



HTTP in Detail completed!

mahnnorchaudary completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
7	176	1

Share on social media
Share your win before starting the next room

[in LinkedIn](#) [WhatsApp](#) [Telegram](#) [X Twitter / X](#) [Facebook](#) [Reddit](#)

[Go to dashboard](#) [Start the next room →](#)

Figure 2 – TryHackMe: “HTTP in Detail” room completion summary (7 tasks, 176 points).

3. Practical Networking Commands

All commands below were executed in Windows PowerShell against the target domain google.com. Each subsection includes the command, the raw output, and a short analysis of what the output reveals.

3.1 ping – Reachability & Latency Test

Command: `ping google.com`. Ping sends ICMP Echo Request packets to the target and measures the round-trip time (RTT) for each reply, confirming that the host is reachable and giving a first impression of network latency.

```
PS C:\Users\Admin> ping google.com

Pinging google.com [2a00:1450:4019:803::200e] with 32 bytes of data:
Reply from 2a00:1450:4019:803::200e: time=75ms
Reply from 2a00:1450:4019:803::200e: time=45ms
Reply from 2a00:1450:4019:803::200e: time=45ms
Reply from 2a00:1450:4019:803::200e: time=45ms

Ping statistics for 2a00:1450:4019:803::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 45ms, Maximum = 75ms, Average = 52ms
```

Figure 3 – Output of ping google.com.

Analysis: All 4 packets sent were received (0% packet loss), confirming google.com (resolved to 2a00:1450:4019:803::200e) is reachable over IPv6. RTT ranged from 45 ms to 75 ms, averaging 52 ms – a healthy latency for a public web service reachable over the internet.

3.2 tracert – Path Discovery

Command: `tracert google.com`. Traceroute maps the sequence of routers (hops) a packet passes through to reach the destination by incrementing the IP Time-To-Live (TTL) value and recording which hop expires the packet at each step.

```
PS C:\Users\Admin> tracert google.com

Tracing route to google.com [2a00:1450:4019:803::200e]
over a maximum of 30 hops:

  1    4 ms     5 ms     4 ms    2407:d000:605:0:d49e:6b22:9cdc:38d8
  2    7 ms     7 ms     7 ms    2404:7000:f002:c5::13
  3    9 ms    10 ms     7 ms    2407:d000:600:704::1
  4    7 ms     6 ms     8 ms    2404:7000:f002:c5::14
  5    7 ms     6 ms     6 ms    2404:7000:f003:2::107
  6   14 ms     9 ms     8 ms    2404:7000:f003:2::85
  7    8 ms    17 ms     7 ms    2404:7000:f001:2::8:12
  8   27 ms    27 ms    27 ms    2404:7000:f001:1::4:16
  9   125 ms   24 ms    30 ms    2404:7000:f001:1::4:6
 10   47 ms   47 ms    47 ms    2001:4860:1:1::b22
 11   45 ms   45 ms    51 ms    fjr02s04-in-x0e.1e100.net [2a00:1450:4019:803::200e]

Trace complete.
```

Figure 4 – Output of tracert google.com (11 hops).

Analysis: The trace completed in 11 hops, moving from the local ISP gateway (hop 1, ~4–5 ms) through several IPv6-addressed backbone/carrier hops, before landing on Google's own network (fjr02s04-in-x0e.1e100.net) at hop 11 with

~45–51 ms latency. The steady, gradually increasing RTT with no timeouts (*) indicates a stable path with no dropped or filtered hops along the route.

3.3 nslookup – DNS Resolution

Command: `nslookup google.com`. This queries the configured DNS resolver directly and returns the IP address(es) associated with a domain name.

```
PS C:\Users\Admin> nslookup google.com
Server: UnKnown
Address: fe80::1

Non-authoritative answer:
Name: google.com
Addresses: 2a00:1450:4019:803::200e
          142.250.187.46
```

Figure 5 – Output of `nslookup google.com`.

Analysis: The non-authoritative answer returned two addresses for `google.com` – an IPv6 address (2a00:1450:4019:803::200e) and an IPv4 address (142.250.187.46) – confirming the domain is dual-stack. “Non-authoritative” indicates the answer came from the resolver's cache rather than directly from `google.com`'s authoritative name servers.

3.4 netstat -ano – Local Connections & Listening Ports

Command: `netstat -ano`. Lists all active TCP/UDP connections and listening ports on the local machine, along with the owning process ID (PID), which is useful for spotting unexpected open ports or services.

```
PS C:\Users\Admin> netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP   0.0.0.0:135              0.0.0.0:0               LISTENING   1856
TCP   0.0.0.0:445              0.0.0.0:0               LISTENING   4
TCP   0.0.0.0:3306             0.0.0.0:0               LISTENING   6388
TCP   0.0.0.0:5040             0.0.0.0:0               LISTENING   10644
TCP   0.0.0.0:7680             0.0.0.0:0               LISTENING   14432
TCP   0.0.0.0:33060            0.0.0.0:0               LISTENING   6388
TCP   0.0.0.0:49664            0.0.0.0:0               LISTENING   1548
TCP   0.0.0.0:49665            0.0.0.0:0               LISTENING   1360
TCP   0.0.0.0:49666            0.0.0.0:0               LISTENING   2332
TCP   0.0.0.0:49667            0.0.0.0:0               LISTENING   3164
TCP   0.0.0.0:49668            0.0.0.0:0               LISTENING   4436
TCP   0.0.0.0:49673            0.0.0.0:0               LISTENING   1500
TCP   127.0.0.1:902            0.0.0.0:0               LISTENING   6336
TCP   127.0.0.1:912            0.0.0.0:0               LISTENING   6336
TCP   127.0.0.1:9222           0.0.0.0:0               LISTENING   25420
TCP   127.0.0.1:11434          0.0.0.0:0               LISTENING   7348
TCP   127.0.0.1:29772          0.0.0.0:0               LISTENING   9900
TCP   127.0.0.1:42804          127.0.0.1:42805         ESTABLISHED 6336
TCP   127.0.0.1:42805          127.0.0.1:42804         ESTABLISHED 6336
TCP   127.0.0.1:49669          127.0.0.1:49670         ESTABLISHED 6388
TCP   127.0.0.1:49670          127.0.0.1:49669         ESTABLISHED 6388
TCP   127.0.0.1:49671          127.0.0.1:49672         ESTABLISHED 6388
TCP   127.0.0.1:49672          127.0.0.1:49671         ESTABLISHED 6388
TCP   192.168.10.1:139         0.0.0.0:0               LISTENING   4
```

Figure 6 – Output of netstat -ano (active connections).

Analysis: The output shows a mix of **LISTENING** ports on 0.0.0.0 (services accepting connections from any interface, e.g. 135/RPC, 445/SMB, 3306/MySQL) and several high, ephemeral ports (49664–49673) used by local applications. A block of **ESTABLISHED** loopback (127.0.0.1) connections around PIDs 6336/6388 indicates local inter-process communication between background applications rather than external traffic – nothing here suggests unexpected external exposure.

4. Subnetting Basics

An IPv4 address is split into a **network portion** and a **host portion**. The subnet mask (or CIDR suffix, e.g. /24) defines how many of the 32 bits belong to the network, leaving the remainder for host addresses within that network.

CIDR notation writes the mask as a slash followed by the number of network bits, e.g. 192.168.1.0/24 means the first 24 bits (3 octets) identify the network and the last 8 bits identify hosts on that network.

4.1 Worked Example: Hosts in a /24 Network

- **Network:** 192.168.1.0/24
- **Subnet mask:** 255.255.255.0
- **Network bits:** 24 → **Host bits:** 32 – 24 = 8
- **Total addresses:** $2^8 = 256$
- **Usable hosts:** $256 - 2 = 254$ (subtract 1 for the network address 192.168.1.0 and 1 for the broadcast address 192.168.1.255)
- **Usable host range:** 192.168.1.1 – 192.168.1.254

Field	Value
Network address	192.168.1.0
Subnet mask	255.255.255.0 (/24)
Broadcast address	192.168.1.255
Usable host range	192.168.1.1 – 192.168.1.254
Total usable hosts	254

Why it matters for reconnaissance: Recognising a /24 quickly tells an analyst the scan scope is 254 live-host addresses – small enough for a full ping sweep or port scan in a reasonable time, unlike a /16 (65,534 hosts) which requires a different scanning strategy.

5. Conclusion

This task built a working foundation in host-reachability testing (ping), path discovery (traceroute), DNS resolution (nslookup) and local connection auditing (netstat), alongside the arithmetic needed to reason about subnet sizes. Both

mandatory TryHackMe rooms were completed and are documented above with proof screenshots. These fundamentals – confirming a host is alive, understanding its network path, resolving its DNS records, and knowing how many addresses a target subnet contains – form the groundwork for the active scanning and enumeration tasks planned next in the internship track.